

Experiment 06:

Initial Commands

```
(student@kali)-[~/Shivam]
$ mkdir forensic_evidence
```

```
(student@kali)-[~/Shivam]
$ ls
forensic_evidence  sample_data
```

```
(student@kali)-[~/Shivam]
$ tar -cvf ./sample_data.tar ./sample_data
./sample_data/
./sample_data/file3.txt
./sample_data/file1.txt
./sample_data/file2.txt
```

Observation:

Created the forensic_evidence directory and compiled the sample files into the sample_data.tar archive using the tar -cvf command.

```
(student@kali)-[~/Shivam]
$ dd if=./sample_data.tar of=./forensic_evidence/evidence.dd bs=4M status=progress
0+1 records in
0+1 records out
10240 bytes (10 kB, 10 KiB) copied, 0.000170245 s, 60.1 MB/s

(student@kali)-[~/Shivam]
$ ls -lh ./forensic_evidence/evidence.dd
-rw-rw-r-- 1 student student 10K Sep 11 11:41 ./forensic_evidence/evidence.dd
```

Observation:

Generated the raw evidence image evidence.dd from sample_data.tar using the dd command and verified its successful creation and file size using ls -lh.

Laboratory Exercise

1. Verification of Bulk Extractor installation.

```
(student@kali)-[~/Shivam]
$ bulk_extractor -V
bulk_extractor 2.1.1
```

Observation:

Executed **bulk_extractor -V** to confirm that version 2.1.1 is installed on the Linux system.

2. Logical image selected for analysis.

```
(student@kali)-[~/Shivam]
$ ls -lh /home/student/Shivam/forensic_evidence/evidence.dd
-rw-rw-r-- 1 student student 10K Sep 11 11:41 /home/student/Shivam/forensic_evidence/evidence.dd
```

Observation:

Verified the presence of the target image file evidence.dd and prepared the destination directory **bulk_output** for storing analysis results.

3. Running Bulk Extractor on the Logical image.

```
(student@kali)-[~/Shivam]
$ bulk_extractor -o bulk_output /home/student/Shivam/forensic_evidence/evidence.dd
bulk_extractor version: 2.1.1
Input file: "/home/student/Shivam/forensic_evidence/evidence.dd"
Output directory: "bulk_output"
Disk Size: 10240
Scanners: aes base64 elf evtx exif facebook find gzip httplogs json kml_carved msxml net ntfsindx ntfs
          accts email gps
Threads: 2
going multi-threaded... ( 2 )
bulk_extractor      Fri Sep 11 11:46:57 2026

available_memory: 3159953408
bytes_queued: 10240
depth0_bytes_queued: 10240
depth0_sbufs_queued: 1
elapsed_time: 0:00:00
estimated_date_completion: 2026-09-11 11:46:56
estimated_time_remaining: n/a
fraction_read: 0.000000 %
max_offset: 0
sbufs_created: 1      All data read; waiting for threads to finish...

sbufs_queued: 1
sbufs_remaining: 1
tasks_queued: 0
thread_count: 2
```

```
Phase 2. Shutting down scanners
Computing final histograms and shutting down...
Phase 3. Generating stats and printing final usage information
All Threads Finished!
Elapsed time: 1.004 sec.
Total MB processed: 0
Overall performance: 0.0102 MBytes/sec 0.0051 (MBytes/sec/thread)
sbufs created: 67
sbufs unaccounted: 0
Time producer spent waiting for scanners to process data: 0:00:00 (0.00 seconds)
Time consumer scanners spent waiting for data from producer: 0:00:00 (0.00 seconds)
Average time each consumer spent waiting for data from producer: 0:00:00 (0.00 seconds)
*** More time spent waiting for workers. You need a faster CPU or more cores for improved performance.
Total email features found: 3
```

Observation:

Ran **bulk_extractor -o bulk_output** on evidence.dd using multi-threading, successfully scanning 10,240 bytes and detecting 3 email features

4. Output directory generated after scanning.

```
(student@kali)-[~/Shivam]
$ ls bulk_output
aes_keys.txt          elf.txt              exif.txt             jpeg.txt
alerts.txt           email_domain_histogram.txt  facebook.txt         json.txt
ccn_histogram.txt     email_histogram.txt  find_histogram.txt   kml_carved.txt
ccn_track2_histogram.txt  email.txt           find.txt             ntfsindx_carved.txt
ccn_track2.txt        ether_histogram_1.txt  gps.txt              ntfslogfile_carved.txt
ccn.txt              ether_histogram.txt   httplogs.txt         ntfsmft_carved.txt
domain_histogram.txt  ether.txt            ip_histogram.txt     ntfsusn_carved.txt
domain.txt           evtX_carved.txt       ip.txt               pii_teamviewer.txt
```

Observation:

Listed the generated contents of bulk_output, showing created artifact files including email.txt, url.txt, ip.txt, and report.xml.

5. Extracted email addresses.

```
(student@kali)-[~/Shivam]
$ cat ./bulk_output/email.txt
# BANNER FILE NOT PROVIDED (-b option)
# BULK_EXTRACTOR-Version: 2.1.1
# Feature-Recorder: email
# Filename: /home/student/Shivam/forensic_evidence/evidence.dd
# Feature-File-Version: 1.1
1024  mansi_ashu@gmail.com      \000\000\000\000\000\000\000\000\000\000\000\000\000\000\000\000\
2048  shivam@gmail.com        \000\000\000\000\000\000\000\000\000\000\000\000\000\000\000\000\
3072  radha@gmail.com         \000\000\000\000\000\000\000\000\000\000\000\000\000\000\000radha
```

Observation:

Displayed the contents of bulk_output/email.txt using cat, revealing extracted email addresses (mansi_ashu@gmail.com, shivam@gmail.com, radha@gmail.com) alongside their byte offsets.

6. Extracted URLs.

```
(student@kali)-[~/Shivam]
$ cat ./bulk_output/url.txt
# BANNER FILE NOT PROVIDED (-b option)
# BULK_EXTRACTOR-Version: 2.1.1
# Feature-Recorder: url
# Filename: /home/student/Shivam/forensic_evidence/evidence.dd
# Feature-File-Version: 1.1
1045  https://java_learning_website_for_her.com      _ashu@gmail.com\012https://java_learn
2065  https://cloudfare.com      hivam@gmail.com\012https://cloudfare.com\012192.168.1.1\012\0
3088  https://github.com      radha@gmail.com\012https://github.com\012192.168.1.2\012\000\
```

Observation:

Displayed **bulk_output/url.txt** using **cat**, showing parsed web addresses, <https://cloudflare.com> , <https://github.com> , extracted from the raw image.

7. Summary report generated by Bulk Extractor.

```
(student@kali)-[~/Shivam]
$ cat ./bulk_output/report.xml
<?xml version='1.0' encoding='UTF-8'?>
<dfxml xmloutputversion='1.0' xmlns:debug='http://afflib.org/bulk_extractor/debug'>
  <metadata
    xmlns='http://afflib.org/bulk_extractor/'
    xmlns:xsi='http://www.w3.org/2001/XMLSchema-instance'
    xmlns:dc='http://purl.org/dc/elements/1.1/'>
    <dc:type>Feature Extraction</dc:type>
  </metadata>
  <creator version='1.0'>
    <program>BULK_EXTRACTOR</program>
    <version>2.1.1</version>
    <build_environment>
      <compiler>14.2.0 (14.2.0)</compiler>
      <CPPFLAGS>-DUTC_OFFSET=+0000 -D_FORTIFY_SOURCE=2 -I/usr/local/include -Wdate-time </CPPFLAGS>
      <CFLAGS>-O2 -O3 -Werror=format-security -Werror=implicit-function-declaration -Wformat -fPIC -fno-ash-protection -fstack-protector-strong -g </CFLAGS>
      <CXXFLAGS>-MD -O2 -O3 -Wall -Wcast-align -Wdisabled-optimization -Weffc++ -Werror=format-security -Wredundant-decls -Wshadow -Wsign-promo -Wwrite-strings -fPIC -fcf-protection -ffile-prefix-map=/build-
      ng -g </CXXFLAGS>
      <LDFLAGS>-Wl,-z,relro -Wl,-z,now -L/usr/local/lib</LDFLAGS>
      <LIBS>-lewf -lstdc++ -lexpat -lz -ldl -lsqlite3 -lcrypto -lgpg-error -lexpat </LIBS>
      <compilation_date>2025-03-14T03:24:33</compilation_date>
      <library name="boost" version="108300"/>
      <library name="libewf" version="20140816"/>
      <library name="sqlite" version="3.46.1" source_id="2024-08-13 09:16:08 c9c2ab54ba1f5f46360f1b4f3">
    </build_environment>
    <execution_environment>
      <cpuid>
        <identification>GenuineIntel</identification>
        <family>0</family>
        <model>1</model>
        <stepping>6</stepping>
        <efamily>0</efamily>
        <emodel>0</emodel>
        <brand>71</brand>
        <clflush_size>808</clflush_size>
        <nproc>110</nproc>
        <apicid>117</apicid>
        <L1_cache_size>2097152</L1_cache_size>
      </cpuid>
    </execution_environment>
  </creator>
</dfxml>
```

Observation:

Displayed the **DFXML execution metadata** and feature summary counts in **bulk_output/report.xml**, confirming total extracted features across various categories like URLs and emails.